

Larsen Bier

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Education

University of California, Los Angeles

Expected Graduation Dec '25

Mathematics, B.S.

GPA: 3.980

Minor in Data Science Engineering

Relevant Coursework: *Probability, Optimization, Random Processes, Linear Algebra, Multivariable Calculus, Differential Equations, Mechanics, Algorithms, Game Theory, Image Processing, Machine Learning, Data Science.*

Professional Experience

Computer Vision Researcher (REU Under NSF SBIR Grant) | EPI Imaging, Palo Alto, CA Jun '25– Sep '25

- Developed statistical algorithms to test and evaluate 3D depth camera calibrations on raw data.
- Trained a ResNet classifier to discriminate between empty and connected cable ports with >95% accuracy.
- Automated simulations of EPI's cameras in Blender to remove human transcription errors in experiments.
- Ran experiments to benchmark the localization quality of RealSense cameras for robotic cable manipulation.

Math Instructor | Olga Radko Endowed Math Circle, UCLA Math Department

Jan '25– Jun '25

- Taught modern college-level math to groups of 20+ highly gifted students.
- Wrote lesson plans and solvable, yet nontrivial problems to sharpen students' proof-writing skills.

Economic Analyst Intern | Sun Valley Economic Development, Ketchum, ID

Jun '24 – Sep '24

- Interfaced Excel storage with Python to automate collection and cleaning of 100% of our Census data.
- Developed and ensembled 7 segmented Excel models to analyze monthly tourism in Sun Valley.
- Crafted visualizations and engineered data from 15+ independent sources to compile [Economic Reports](#).

Data Science Intern | Chipotle Mexican Grill, Inc., Newport Beach, CA

Jun '23 – Aug '23

- Leveraged multivariate linear regression in Python to analyze the financial impact of precipitation on sales.
- Developed a scalable pipeline to query millions of rows of data from 500+ stores in SQL.
- Cleaned and documented 3,000+ lines of employees' code to enhance its clarity and usability.

Leadership and Extracurricular Experience

Deep Learning Engineer | DataRes, UCLA

Jan '24 – Jun '25

- Led a 10-week team project implementing diffusion-based generative AI for image denoising.
- Coded a deep reinforcement learning environment for Connect4 to support a computer agent.
- Employed graph theoretic algorithms to publish an [article](#) on the strongest networks in Hollywood.
- Created a deep convolutional neural network for image classification on a dataset of 100 categories.

Skills

Coding: Python, SQL, MATLAB, C++.

Technologies: LaTeX, Robot Operating System (ROS), Git, Command Line, Blender, Excel, PowerPoint, MS Office Suite.

Technical Skills: Image Processing, Computer Vision, Pytorch, Scikit-Learn, Pandas, Matplotlib, Data Visualization/Analysis, Data Engineering, Machine Learning, Statistical Analysis, Optimization.

Other: Spanish (Fluent), Teaching.